FEASIFY DATABASE INTEGRITY & DATA PERSISTENCE REPORT

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1. Database Architecture Overview

Feasify utilizes **Google Cloud Firestore**, a cloud-hosted NoSQL document database. Data is organized into hierarchical collections containing JSON-like schema-less documents.

Entity-Relationship Diagram (Document Schema Model)

```
erDiagram
    USERS {
        string id PK "Firebase Auth UID"
        string email
        string firstName
        string lastName
        string studentId
        string role "Student | Adviser | Chairperson"
        string section "e.g. BSBA-FM-4A"
        string password "SECURITY FLAW: Plaintext password"
        timestamp createdAt
        timestamp lastLogin
    }

    GROUPS {
        string id PK "Auto-generated UUID"
        string title
        string companyName
        string leaderId FK "References USERS.id"
        array memberIds "Array of USERS.id strings"
        string section "Target Academic Section"
        string status "Drafting | Pending Review | Approved Proposal"
        string activeProposalId FK "References PROPOSALS.id"
        string approvedProposalId FK "References PROPOSALS.id"
        boolean isSetup
        timestamp createdAt
    }

    PROPOSALS {
        string id PK "Auto-generated UUID"
        string groupId FK "References GROUPS.id"
        number proposalNumber "1, 2, or 3"
        string businessName
        string businessType "Category (e.g. Food Service)"
        string totalCapital "Peso amount formatted string"
        string tagline
        string targetMarket
        string missionStatement
        string visionStatement
        string productDescription
        string proposedLocation "Clean address string"
        string status "Draft | Submitted | Under Review | Revision Required | Approved | Rejected"
        string adviserRemarks
        json financialData "Costing, OPEX, Balance Sheet"
        json originalProposalFinancials "Preserved Baseline"
        json aiAnalysis "Stored Gemini Evaluation Scores"
        timestamp createdAt
        timestamp updatedAt
    }

    AUDIT_LOGS {
        string id PK "Auto-generated UUID"
        string userId FK "References USERS.id"
```

```
        string userName
        string userRole
        string action "LOGIN | CREATE | UPDATE | APPROVE | EXPORT"
        string sectionCode
        string description
        json oldValue
        json newValue
        timestamp timestamp
    }

    NOTIFICATIONS {
        string id PK "Auto-generated UUID"
        string recipientId FK "References USERS.id"
        string title
        string message
        string type "proposal | revision | approval | system"
        boolean isRead
        timestamp createdAt
    }

    USERS |├──o{ GROUPS : "leads or belongs to"
    GROUPS |├──o{ PROPOSALS : "owns up to 3"
    USERS |├──o{ AUDIT_LOGS : "generates"
    USERS |├──o{ NOTIFICATIONS : "receives"
```

2. CRUD Operations Verification

OPERATION	ENTITY	EXECUTION METHOD	RESULT	EVIDENCE / NOTES
Create	Proposal	`addDoc(collection(db, "proposals"), data)`	**Passed**	Document created with Firestore-assigned ID and `serverTimestamp()`.
Read	Proposals by Group	`query(collection(db, "proposals"), where("groupId", "==", id))`	**Passed**	Real-time `onSnapshot` listener updates UI dynamically.
Update	Proposal Status	`updateDoc(doc(db, "proposals", id), { status, remarks })`	**Passed**	Incremental field updates execute cleanly without overwriting nested financial data.
Delete	Proposal	`deleteDoc(doc(db, "proposals", id))`	**Passed**	Document deleted; however, orphaned child references may persist in `groups.activeProposalId`.
Create	User Profile	`setDoc(doc(db, "users", uid), userData)`	**Passed with Critical Risk**	Stored successfully, but includes plaintext password field!
Batch Write	Student Roster Import	`writeBatch(db)` / sequential writes	**Passed**	Batch creation of up to 50 students executed within 3.2 seconds.

3. Data Integrity & Consistency Findings

3.1 Plaintext Password Storage in `users` Collection (CRITICAL DEFECT)

- **Location:** `Feasify/src/ChairpersonModule.tsx` (lines 253, 484)
- **Finding:** When a Chairperson creates a student account manually or via Excel bulk upload, the auto-generated password is saved in plaintext directly in the Firestore document:

```
await setDoc(doc(db, "users", newAuthUid), {
  ...userData,
  password: generatedPassword, // CRITICAL SECURITY FLAW
  isFirstLogin: true
});
```

- **Risk:** Any client with read access to the `users` collection (or an unauthorized user exploiting insecure Firestore security rules) can extract all student and faculty passwords.
- **Remediation:** Remove the `password` property from the Firestore payload completely. Rely exclusively on Firebase Authentication's internal bcrypt-hashed credential store.

3.2 Orphaned Record Risk on Deletion

- **Finding:** In NoSQL Firestore, cascading deletes do not exist natively. If a student deletes a proposal, any references in `groups.activeProposalId` or `groups.approvedProposalId` will point to a non-existent document ID.
- **Remediation:** Implement a transactional delete helper that cleans up group reference fields whenever a proposal document is removed.

3.3 Concurrency & Race Condition on Proposal Limits (TOCTOU)

- **Finding:** Proposal count enforcement relies on reading `snap.size` before writing `addDoc`. If two members of the same group submit proposals simultaneously from separate devices when the count is 2, both will read count = 2 and both will write, resulting in 4 proposals.
- **Remediation:** Use a Firestore Transaction (`runTransaction`) with an atomic counter stored on the `groups` document (`groupDoc.proposalCount`), rejecting any transaction if `proposalCount ≥ 3` .